

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2128

Waterford Power Llc 230 kV
October 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2128. The Interconnection Customer proposes a 120.8 MW solar photovoltaic facility interconnecting at the Waterford Power Llc 230 kV substation in the Ohio area. The request was received on July 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 653 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,699,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Brooks Hill - Ore 3 Inc 230 KV	Tidd Substation - Muskingum Riv 500 KV	149.3	491	13.8
Brooks Hill - Myrtle-Date 230 KV	Tidd - Muskingum River 500 KV	146.3	518	13.8
Oreg 3 Inc 345-230 kV Transformer Bank	Tidd - Muskingum River 500 KV	140.9	361	9.9
Springhill - W Bellaire 230 KV	Tidd - Muskingum Riv 500 KV	135.2	495	8.8
Mine Rd - W Bellaire Substation 230 KV	Tidd - Muskingum Riv 500 KV Line	126.4	684	10.2
Brentwood - Mine Rd 230 KV	Tidd - Muskingum Riv 500 KV L	125.9	695	10.2
Brentwood - Muskingum River 230 KV	Tidd - Muskingum River 500 KV	125.3	707	10.2
Oreg 3 Inc Substation 230 kV/138 kV Auto	Oreg 3 Inc Substation 345/230kV Autotransformer	118.7	204	5.5
Muskingum River - Waterford Power Llc 230 KV	Myrtle-Date - Waterford Power Llc 230 KV Line	104.6	555	62.4
Springhill - Lerdo 230 KV L	Lerdo - Tidd 500 KV Line	104.3	472	5.6
Filer 500 kV/345 kV TX	Tidd - Lerdo 500 KV	101.4	1,238	9.5
Waterford Power Llc - Myrtle-Date 230 KV line	Tidd - Muskingum River 500 KV	100.1	1,045	31.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Oreg 3 Inc - Brooks Hill 230 KV	Rebuild line with higher-capacity conductor	\$10,744,000	11% (\$1,173,000)
Brooks Hill - Myrtle-Date 230 KV	Rebuild line with higher-capacity conductor	\$38,805,000	11% (\$4,235,000)
Oreg 3 Inc 345/230 kV XFMR	Replace transformer with higher-rated unit	\$9,280,000	11% (\$1,058,000)
Springhill - W Bellaire 230 KV	Rebuild line with higher-capacity conductor	\$23,157,000	9% (\$2,112,000)
Mine Rd - W Bellaire 230 KV	Rebuild line with higher-capacity conductor	\$39,614,000	10% (\$4,057,000)
Brentwood - Mine Rd 230 KV	Rebuild line with higher-capacity conductor	\$32,635,000	10% (\$3,343,000)
Brentwood - Muskingum River 230 KV	Rebuild line with higher-capacity conductor	\$52,356,000	10% (\$5,362,000)
Oreg 3 Inc 230/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$9,789,000	14% (\$1,357,000)
Muskingum River - Waterford Power Llc 230 KV	Reconductor line	\$3,365,000	15% (\$502,000)
Springhill - Lerdo 230 KV L	Reconductor line	\$18,651,000	10% (\$1,873,000)
Filer 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$26,223,000	3% (\$872,000)
Waterford Power Llc - Myrtle-Date 230 KV	Reconductor line	\$9,915,000	8% (\$755,000)
	Total allocated to project		\$26,699,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.